

CS & IT ENGINEERING



Computer Network

IPv4 Address

Lecture No. – 07



By - Abhishek Sir



Recap of Previous Lecture



Topic

CIDR

Topic

Routing Table



Topics to be Covered



Topic

Routing Table



ABOUT ME



Hello, I'm **Abhishek**

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[MCQ]

[GATE-2020][2 Mark]



#Q. An organization requires a range of IP address to assign one to each of its 1500 computers. The organization has approached an Internet Service Provider (ISP) for this task. The ISP uses CIDR and serves the requests from the available IP address space "202.61.0.0 / 17". The ISP wants to assign an address space to the organization which will minimize the number of routing entries in the ISP's router using route aggregation. Which of the following address spaces are potential candidates from which the ISP can allot any one of the organization?

~~I. 202.61.84.0 / 21~~

84 → 01010100

✓ II. 202.61.104.0 / 21

104 → 01101000

✓ III. 202.61.64.0 / 21

64 → 01000000

~~IV. 202.61.144.0 / 21~~

144 → 10010000

A I and II only

✓ B II and III only

C III and IV only

D I and IV only

Ans: B

Available IP Address space = 202.61.0.0 / 17

202.61.0.0 ————— / 17
 17 bit prefix 15 bit Host IP

Number of hosts = 1500 hosts
 [for IP address requested]

Number of bits for host id in allocated chunk = $\lceil \log_2(\text{Number of hosts}) \rceil$ bits
 = $\lceil \log_2(1500) \rceil = 11$ bits

Allocated IP Address space =

202.61.0 $b_1 b_2 b_3 b_4 000$. 00000000000 / 21
 21 bit prefix 11 bit host ID

[MCQ]

[GATE-2012][2 Mark]



#Q. An Internet Service Provider (ISP) has the following chunk of CIDR-based IP addresses available with it : "245 . 248 . 128 . 0 / 20". The ISP wants to give half of this chunk of addresses to Organization A, and a quarter to Organization B, while retaining the remaining with itself. Which of the following is a valid allocation of addresses to A and B?

☒ A 245 . 248 . 136 . 0 / 21 and 245 . 248 . 128 . 0 / 22

☐ B 245 . 248 . 128 . 0 / 21 and 245 . 248 . 128 . 0 / 22

☐ C 245 . 248 . 132 . 0 / 22 and 245 . 248 . 132 . 0 / 21

☐ D 245 . 248 . 136 . 0 / 24 and 245 . 248 . 132 . 0 / 21

A $\rightarrow$ /21

B $\rightarrow$ /22

ISP $\rightarrow$ /22

Ans: A

CIDR based available IP address space : 245 . 248 . 128 . 0 / 20

245 . 248 . 1000 0000 . 00000000 / 20
 20 bit prefix 12 bit host ID

Org.-1 Network Address :

245 . 248 . 1000 0000 . 00000000 / 21
 21 bit prefix 11 bit host ID

Org.-2 Network Address :

245 . 248 . 1000 1000 . 00000000 / 22
 22 bit prefix 10 bit host ID

ISP available Address Space :

245 . 248 . 1000 1100 . 00000000 / 22
 22 bit prefix 10 bit host ID

CIDR based available IP address space : 245 . 248 . 128 . 0 / 20

245 . 248 . 1 0 0 0 _ _ _ _ . _ _ _ _ _ _ _ _ _ _ / 20

Org.-1 Network Address :

245 . 248 . 1 0 0 0 | 0 0 0 . 0 0 0 0 0 0 0 0 0 0 / 21
 11 bit host ID

Org.-2 Network Address :

245 . 248 . 1 0 0 0 | 0 0 0 . 0 0 0 0 0 0 0 0 0 0 / 22
 10 bit host ID

ISP available Address Space :

245 . 248 . 1 0 0 0 | 0 1 0 0 . 0 0 0 0 0 0 0 0 0 0 / 22
 10 bit host ID

Org.-1 Net. Add. : 245 . 248 . 128 . 0 / 21
 Org.-2 Net. Add. : 245 . 248 . 136 . 0 / 22
 ISP Remaining : 245 . 248 . 140 . 0 / 22

✓
 Org.-1 Net. Add. : 245 . 248 . 136 . 0 / 21
 Org.-2 Net. Add. : 245 . 248 . 128 . 0 / 22
 ISP Remaining : 245 . 248 . 132 . 0 / 22

Org.-1 Net. Add. : 245 . 248 . 128 . 0 / 21
 Org.-2 Net. Add. : 245 . 248 . 140 . 0 / 22
 ISP Remaining : 245 . 248 . 136 . 0 / 22

Org.-1 Net. Add. : 245 . 248 . 136 . 0 / 21
 Org.-2 Net. Add. : 245 . 248 . 132 . 0 / 22
 ISP Remaining : 245 . 248 . 128 . 0 / 22

[MCQ]

[GATE-2024][2 Mark]



#Q. Which one of the following CIDR prefixes exactly represents the range of IP addresses 10.12.2.0 to 10.12.3.255?

☒ A 10.12.2.0/23

☐ B 10.12.2.0/24

☒ C 10.12.0.0/22

☐ D 10.12.2.0/22

16

Ans: A

10.12.2.0 : 10.12.00000001 0.00000000

10.12.3.255 : 10.12.00000001 1.11111111

CIDR Prefixes : 10.12.00000001 0.00000000 / 23

Network Address :
 ↳ 10.12.2.0 / 23

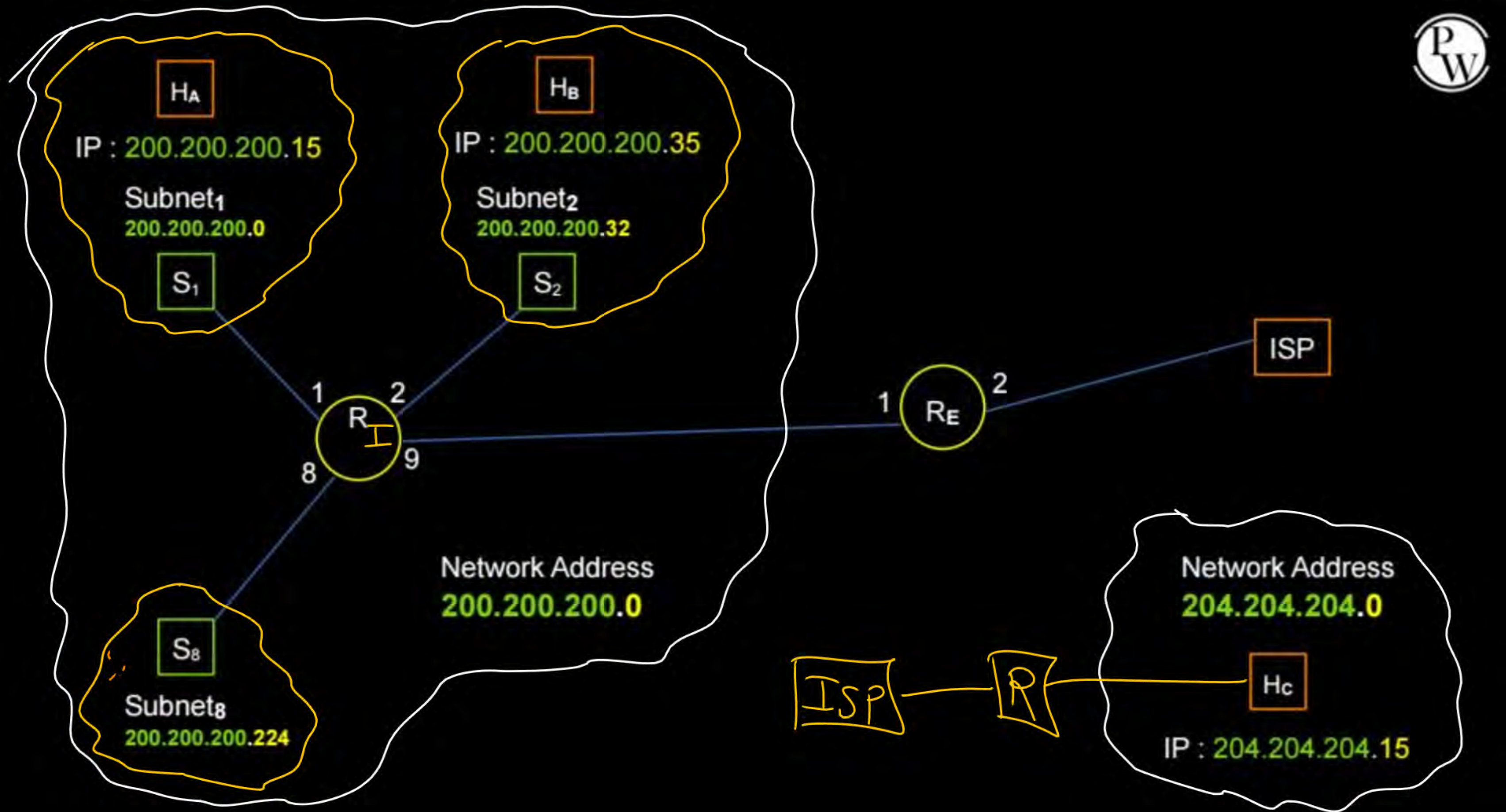
Class C Network :-

Network Address	: 200 . 200 . 200 . 0
Default Netmask	: 255 . 255 . 255 . 0

After 3-bit subnetting :-

Subnet Mask	: 255 . 255 . 255 . 224
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1 st Subnet Address	: 200 . 200 . 200 . 0
2 nd Subnet Address	: 200 . 200 . 200 . 32
3 rd Subnet Address	: 200 . 200 . 200 . 64
4 th Subnet Address	: 200 . 200 . 200 . 96
5 th Subnet Address	: 200 . 200 . 200 . 128
6 th Subnet Address	: 200 . 200 . 200 . 160
7 th Subnet Address	: 200 . 200 . 200 . 192
8 th Subnet Address	: 200 . 200 . 200 . 224





Topic : Forwarding Table



Router (**R_E**) forwarding table

Network(Address)	Network Mask	Interface ID	Next Hop
<u>200</u> . 200 . 200 . 0	255. 255 . 255 . 0	1	R _I
Default		2	ISP



Topic : Forwarding Table



Router (R_1) forwarding table

Subnet(Address)	Subnet(Mask)	Interface ID	Next Hop
200 . 200 . 200 . 0	255 . 255 . 255 . 224	1	On Link
[200 . 200 . 200 . 32]	255 . 255 . 255 . 224	2	[On Link]
200 . 200 . 200 . 64	255 . 255 . 255 . 224	3	On Link
200 . 200 . 200 . 224	255 . 255 . 255 . 224	8	On Link
Default		9	R_E



Topic : Forwarding Table



CASE I:

Source IP Address : 200 . 200 . 200 . 15

HA

Destination IP Address : 200 . 200 . 200 . 35

HB

35 → 00100011
224 → 11100000
32 ← 00100000

Router	Interface ID	Next Hop
<u>R₁</u>	2	ON-LINK

Destination IP Address : 200 . 200 . 200 . 35

Subnet Mask : 255 . 255 . 255 . 224

Address ← 200 . 200 . 200 . 32



Topic : Forwarding Table



CASE II :

Source IP Address : 200.200.200.35 H_B

Destination IP Address : 204.204.204.15 H_C

Router	Interface ID	Next Hop
<u>R_I</u>	9	R _E
R _E	2	ISP

R_E

IP Address : 204.204.204.15

Subnet Mask : 255.255.255.224

Address ← 204.204.204.

R_E

IP Address : 204.204.204.15

Net Mask : 255.255.255.0

Address ← 204.204.204.0



Topic : Forwarding Table



CASE III :

Source IP Address : 204.204.204.15 H_C

Destination IP Address : 200.200.200.35 H_B

Router	Interface ID	Next Hop
<u>R_E</u>	1	R _I
<u>R_I</u>	2	ON-LINK

R_E

IP Address : 200.200.200.35

Net Mask : 255.255.255.0

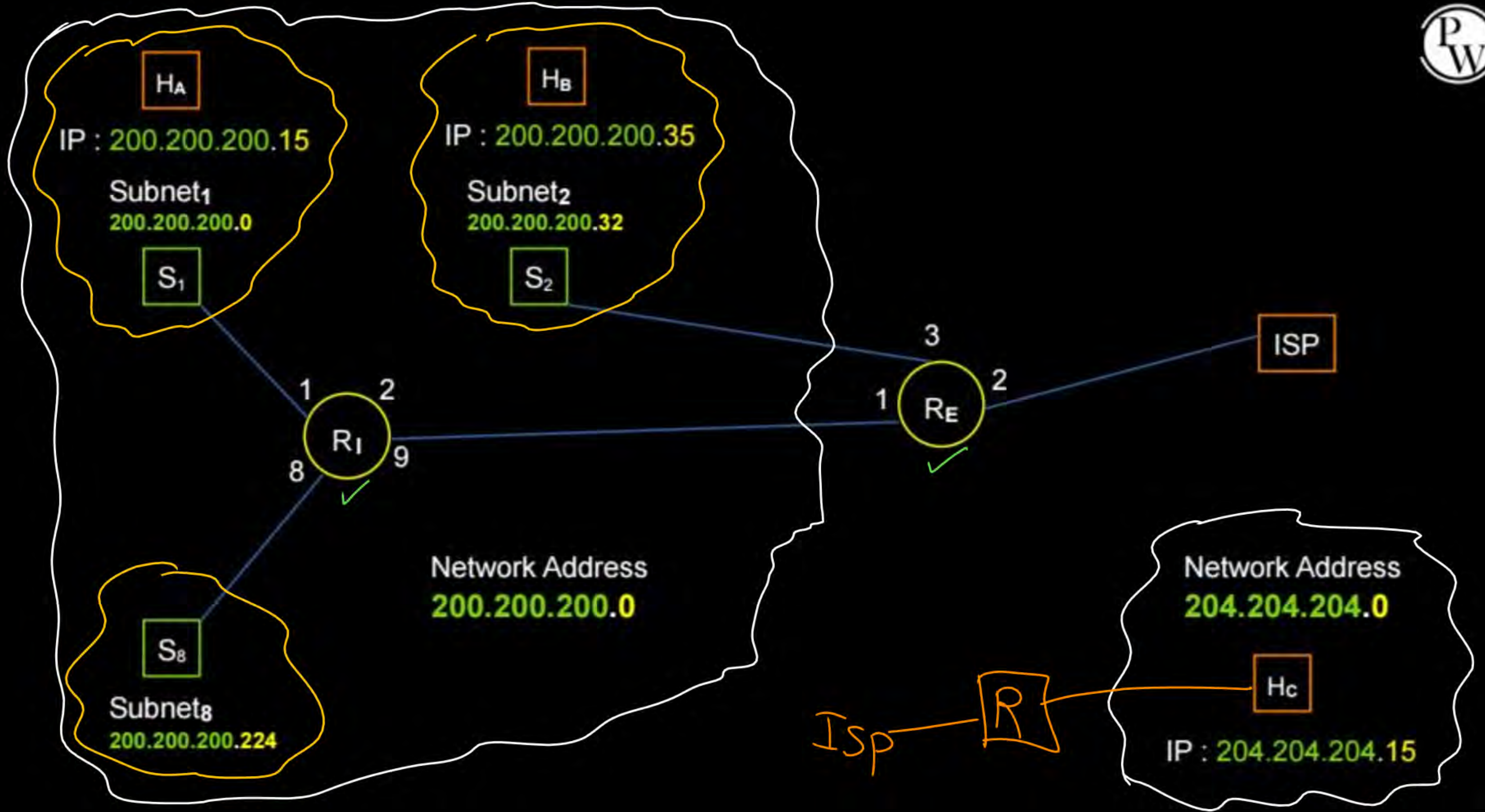
Address ← 200.200.200.0

R_I

IP Address : 200.200.200.35

Subnet Mask : 255.255.255.224

Address ← 200.200.200.32





Topic : Forwarding Table



Router (**R_E**) Updated forwarding table

Network (Address)	Network Mask	Interface ID	Next Hop
→ 200 . 200 . 200 . 0	255 . 255 . 255 . 0	1	R _I
⇒ 200 . 200 . 200 . 32	255 . 255 . 255 . 224	<u>3</u>	<u>On Link</u>
→ 0 . 0 . 0 . 0 [Default]	0 . 0 . 0 . 0	2	ISP



Topic : Forwarding Table



Router (R_1) Updated forwarding table

Subnet Address	Subnet Mask	Interface ID	Next Hop
200 . 200 . 200 . 0	255 . 255 . 255 . 224	1	On Link
200 . 200 . 200 . 64	255 . 255 . 255 . 224	3	On Link
⋮	⋮		
200 . 200 . 200 . 224	255 . 255 . 255 . 224	8	On Link
Default		9	R_E



Topic : Forwarding Table



CASE IV ;

Source IP Address : 204 . 204 . 204 . 15 HC

Destination IP Address : 200 . 200 . 200 . 35 H-B

Router	Interface ID	Next Hop
<u>RE</u>	3	ON-LINK

IP Address : 200 . 200 . 200 . 35
Net Mask : 255 . 255 . 255 . 0

Address ← 200 . 200 . 200 . 0

IP Address : 200 . 200 . 200 . 35
Subnet Mask : 255 . 255 . 255 . 224

Address ← 200 . 200 . 200 . 32

IP Address : 200 . 200 . 200 . 35
Mask : 0 . 0 . 0 . 0

0 . 0 . 0 . 0



Topic : Longest Prefix Match



→ Router chooses more specific option over generic one

→ Router chooses the matched entry in which more number of ones in netmask

* Default Entry in router's routing table is optional

CS-23, CS-24, CS-25

* when Routing Table does not have default Entry.

⇒ If no any present entry matches for an IP Packet.

then Router discard the packet and send ICMP Error Message "Destination Network Unreachable" to source host

[MCQ]

IIT-KGP

[GATE-IT-2006] [2 Mark]



#Q. A router uses following routing table :

Destination	Mask	Interface
144 . 16 . 0 . 0	255 . 255 . 0 . 0	Eth0
144 . 16 . 64 . 0	255 . 255 . 224 . 0	Eth1
<u>144 . 16 . 68 . 0</u>	255 . 255 . 255 . 0	Eth2 ✓
<u>144 . 16 . 68 . 64</u>	255 . 255 . 255 . 224	Eth3 ✗

A packet bearing a destination address 144 . 16 . 68 . 117 arrive at the router.
On which interface will it be forwarded ?

A Eth0

B Eth1

✓ C Eth2

D Eth3

Ans: C

144 . 16 . 68 . 117

255 . 255 . 255 . 224

144 . 16 . 68 . 96

117 → 01110101
 224 → 11100000

 96 ← 01100000

144 . 16 . 68 . 117

255 . 255 . 255 . 0

144 . 16 . 68 . 0

[MCQ]

H.W.

IIT-D

[GATE-CS-2004] [2 Mark]



#Q. The routing table of a router shown below :

Destination	Subnet Mask	Interface
128 . 75 . 43 . 0	255 . 255 . 255 . 0	Eth0
128 . 75 . 43 . 0	255 . 255 . 255 . 128	Eth1
192 . 12 . 17 . 5	255 . 255 . 255 . 255	Eth3
<u>Default</u> (0.0.0.0) (0.0.0.0)		Eth2

On which interfaces will the router forward packets addressed to destinations 128 . 75 . 43 . 16 and 192 . 12 . 17 . 10 respectively ?

- | | | | |
|-------------------------|---------------|-------------------------|---------------|
| ☐ A | Eth1 and Eth2 | ☐ B | Eth0 and Eth2 |
| ☐ C | Eth0 and Eth3 | ☐ D | Eth1 and Eth3 |

[NAT]

IIT-K, H.W.

[GATE-2023][2 Mark]



#Q. The forwarding table of a router is shown below.

Subnet Number	Subnet Mask	Interface ID
200.150.0.0	255.255.0.0	1
200.150.64.0	255.255.224.0	2
200.150.68.0	255.255.255.0	3
200.150.68.64	255.255.255.240	4
Default		0

A packet addressed to a destination address 200.150.68.118 arrives at the router. It will be forwarded to the interface with ID _____.

#Q. An IP router implementing Classless Inter-domain Routing (CIDR) receives a packet with address 131 . 23 . 151 . 76 . The router's routing table has the following entries :

Prefix	Output Interface
131 . 16 . 0 . 0 / 12	3
131 . 28 . 0 . 0 / 14	5
131 . <u>19</u> . 0 . 0 / <u>16</u>	2
<u>131</u> . <u>22</u> . 0 . 0 / <u>15</u>	1

The identifier of the output interface on which this packet will be forwarded is _____.

Ans = 1

131. 23 . 151 . 76

255 . 255 . 0 . 0

131. 23 . 0 . 0 X

131. 23 . 151 . 76

255 . 254 . 0 . 0

131 . 22 . 0 . 0

23 → 00010111

254 → 11111110

22 ← 00010110

[MCQ]

IIT-R, H.W.

[GATE-2025][2 Mark]

#Q. A packet with the destination IP address 145.36.109.70 arrives at a router whose routing table is shown. Which interface will the packet be forwarded to?

Subnet	Subnet Mask (in CIDR notation)	Interface
145.36.0.0	/16	E1
145.36.128.0	/17	E2
145.36.64.0	/18	E3
145.36.225.0	/24	E4
Default	--	E5

A

E3

B

E1

C

E2

D

E5



2 mins Summary



Topic

Routing Table

→ V.V.Imp.



THANK - YOU

